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Nguyen Khanh Toan

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Security student with a solid foundation in digital forensics, reverse engineering, and OS internals. Proven agility in artifact exploitation, and low-level system programming through advanced labs and CTFs. Aspiring to leverage an investigative mindset to excel as a Mobile Security Engineer Intern – Forensics.

Education

VNU-HCM University of Science (HCMUS) **B.Sc. Information Technology, Major in Cybersecurity** **Aug 2024 – Expected 2028**

- Second-Year Student (Class of 2024); Ho Chi Minh City, Vietnam
- **GPA: 3.67/4.0**
- Relevant Coursework: Computer Networks, Operating Systems, Network Security Basics, Systems Architecture

Nguyen Dinh Chieu High School for the Gifted **Major in Mathematics** **Graduated Jun 2024**

- Admitted with maximum scores (10/10) in Mathematics and English entrance exams; 2nd Prize in Provincial Physics Competition
- 12 consecutive years of Excellence; Grade 12 GPA: 9.4/10 (Physics: 9.5, English: 9.7); Dong Thap, Vietnam

Cybersecurity Projects

Centralized Wazuh SIEM & Detection Engineering Lab [GitHub](#) **Independent Project**

- Core Tech: Wazuh SIEM/XDR, Docker, Kali Linux, Microsoft Sysmon, Telegram Webhooks
- Designed and deployed a centralized SOC home lab for **digital forensics investigation**, collecting endpoint telemetry to reconstruct **attack timelines** and perform **root cause analysis** of security incidents.
- Orchestrated the Wazuh Stack (Indexer, Dashboard, Manager) via Docker Compose; configured Windows Sysmon XML rules for **artifact collection**; authored custom Regex rules for obfuscated PowerShell and LFI detection; implemented Telegram bot alerts for severity ≥ 10 .
- Simulated brute-force events (T1110) via Hydra; integrated Active Response to auto-block rogue IPs for 600 seconds, enabling **incident response automation** and minimizing attack surface for forensic analysis.

Secure FTP Client with Real-Time Virus Scanning [GitHub](#) **Computer Networks Lab**

- Core Tech: Python, Socket Programming, ClamAV/ClamD, FTP Protocol, FileZilla Server
- Developed a multi-component secure file transfer infrastructure with real-time **malware analysis** capabilities to address vulnerabilities in legacy FTP transfers.
- Built an interactive Python FTP Client (PASV mode, multi-file transfers) coupled with a clamav_agent.py listener for **threat detection**; configured ClamD to return **malware signatures** dynamically; enforced network rules via Windows Firewall.
- Guaranteed 100% inspection of incoming files before storage, effectively preventing EICAR test malware from infecting server paths through **behavioral analysis** and signature-based detection.

TravelTogether – Smart Tourism Platform (Security Module) [GitHub](#) **Collaborative Project**

- Core Tech: FastAPI, PostgreSQL (Supabase), Firebase Admin SDK, JWT, Encryption, Flutter
- Architected backend security controls, authentication workflows, and an automated emergency fallback system to protect user metadata and privacy through **secure data handling**.
- Enforced JWT-based session handling via Firebase Admin SDK; engineered an autonomous Dual-PIN security module monitoring user safety timestamps, auto-sending encrypted emergency GPS to kin emails on 36-hour inactivity.
- Eliminated identity-spoofing vectors and established a privacy-first backend architecture verified through rigorous **security testing** and vulnerability assessment.

Capture The Flag (CTF) Journey & Writeup Repository [GitHub](#) **Active Practice**

- Core Tech: Linux CLI, GDB, Ghidra, Hex Dump Parsers, Reverse Engineering Tools
- Analyzed and documented structural exploitation mechanisms on public platforms to compile educational **digital forensics** content and **artifact analysis** methodologies.
- Solved multi-tier challenges on picoCTF; performed **binary analysis** and **memory forensics** via GDB; extracted **IoCs**; developed Python scripts to analyze **hex dumps** and isolate hidden payloads in **slack-space anomalies**.
- Maintained an active, peer-reviewed repository with step-by-step **forensic investigation** pathways and automated scripts across primary cybersecurity domains, demonstrating proficiency in **reverse engineering** and **malware behavior analysis**.

Technical Skills

Digital Forensics & Incident Response Triaging, Incident Detection	Artifact Analysis, Attack Timeline Reconstruction, Root Cause Analysis, IoC Extraction, Memory Forensics, Log
Reverse Engineering & Malware Analysis Payload Extraction	GDB, Ghidra, Hex Dump Processing, Binary Analysis, Malware Behavior Analysis, Slack-Space Investigation,
Network & Endpoint Security SIEM/XDR, Microsoft Sentinel	Wireshark, Nmap, FTP/TLS, Port-Forwarding, Firewall Rule Management, Sysmon Telemetry, Wazuh
OS Internals & Hardening	Kali Linux, Ubuntu, Windows Server, xv6 Kernel Customization, System Programming, Memory Management
Programming & Scripting	Python (forensics scripts, log parsing), C/C++ (system programming, binary analysis), Bash Scripting, SQL
Security Frameworks	MITRE ATT&CK, OWASP Top 10, Threat Hunting, Phishing Investigation, Cryptography Basics

Languages	
English	Advanced
Vietnamese	Native